

Aryl Hydrocarbon Receptor (AHR)-Active Pharmaceuticals Are Selective AHR Modulators in MDA-MB-468 and BT474 Breast Cancer Cells

Leflunomide, flutamide, nimodipine, mexiletine, sulindac, tranilast, 4-hydroxytamoxifen, and omeprazole are pharmaceuticals previously characterized as aryl hydrocarbon receptor (AHR) agonists in various cell lines and animal models. In this study, the eight AHR-active pharmaceuticals were investigated in highly aggressive aryl hydrocarbon (Ah)-responsive BT474 and MDA-MB-468 breast cancer cell lines, and their effects on AHR protein, CYP1A1 (protein and mRNA), CYP1B1 (mRNA), and cell migration were determined. 2,3,7,8-Tetrachlorodibenzo-p-dioxin (TCDD) was used as a positive control. The AHR agonist activities of the pharmaceuticals depended on structure, response, and cell context. Most compounds induced one or more AHR-mediated responses in BT474 cells, whereas in Ah-responsive MDA-MB-468 cells effects of the AHR-active pharmaceuticals were highly variable. 4-Hydroxytamoxifen, mexiletine, and tranilast did not induce CYP1A1 in MDA-MB-468 cells; moreover, in combination with TCDD, mexiletine was a potent AHR antagonist, tranilast was a partial antagonist, and 4-hydroxytamoxifen also exhibited some AHR antagonist activity. Omeprazole and, to a lesser extent, sulindac and leflunomide were full and partial AHR agonists, respectively, in both breast cancer cell lines. These data indicate that the AHR-active pharmaceuticals are selective AHR modulators, and applications of these drugs for targeting the AHR must be confirmed by studies using the most relevant cell context.